

Campus Placement Prediction Analysis Report

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Introduction

Campus placement or campus recruiting is a program conducted within educational institutions to help students nearing the completion of their studies acquire jobs. In this type of program, educational institutions partner with corporations that wish to recruit from the student population. Campus recruiting is a method for medium- and large-sized companies with high recruiting needs to hire people who meet their recruiting needs with ease.

Understanding Problem Statement

XYZ University wants to build a machine learning model to know whether students will get placed or not based on their academic background and work experience. So that they can provide special attention and help them get a job, the dataset can be treated as a classification or regression problem. To perform Exploratory Data Analysis to check what factors influence the placement of a candidate.

About Dataset

The dataset had 14 variables in total whose description is given below:

Variable	Description
sl_no	Serial Number
gender	Student Gender - 'M' for Male and 'F' for female
ssc_p	SSC (10th) Percentage
ssc_b	SSC Board of Education - Central or Others
hsc_p	HSC (12th) Percentage
hsc_b	HSC Board of Education - Central or Others
hsc_s	Specialization in HSC
degree_p	Degree Percentage
degree_t	Under Graduation (Degree type) -field of Degree Education
workex	Work Experience

etest_p	Employability test percentage (conducted by college)
specialisation	Post Graduation (MBA) - Specialisation
mba_p	MBA percentage
status	Status of placement - placed/not placed (target variable)

Essential Python Libraries

```
%matplotlib notebook
%matplotlib inline
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import pandas as pd
import warnings
warnings.filterwarnings('ignore')
```

These are some of the few necessary libraries but as the analysis goes on other libraries are further unboxed.

Import Datasets

Importing the dataset:

```
#Loading the single csv file to a variable named 'placement'
placement=pd.read_csv("../input/factors-affecting-campus-placement/Placement_Data_Full_Class.csv")
```

Data Size:

```
data.shape
```

```
(215, 14)
```

Data Preprocessing

Dropping the unwanted column - sl_no

```
data = data.drop(['sl_no'], axis=1)
```

Check for null values and the data types of the variables using the info() method

```
data.info()
```

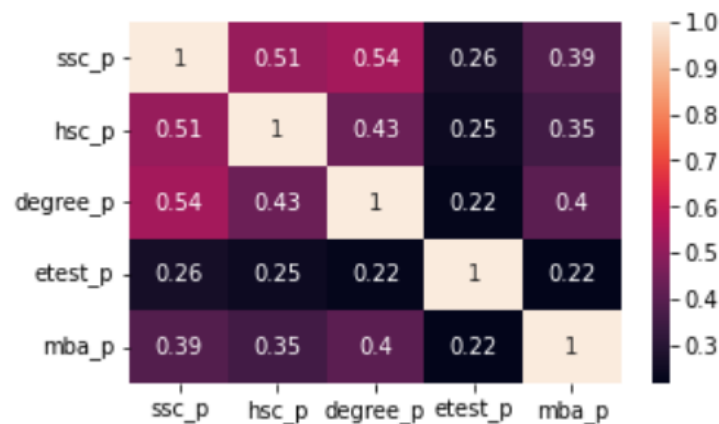
#	Column	Non-Null Count	Dtype
0	gender	215 non-null	object
1	ssc_p	215 non-null	float64
2	ssc_b	215 non-null	object
3	hsc_p	215 non-null	float64
4	hsc_b	215 non-null	object
5	hsc_s	215 non-null	object
6	degree_p	215 non-null	float64
7	degree_t	215 non-null	object
8	workex	215 non-null	object
9	etest_p	215 non-null	float64
10	specialisation	215 non-null	object
11	mba_p	215 non-null	float64
12	status	215 non-null	object

- There were no null values in any columns.
- **Categorical Columns:** gender (M/F), ssc_b (Central/Others), hsc_b (Others/Central), hsc_s (Commerce/Science/Arts), degree_t (Comm&Mgmt/Sci&Tech/Others), workex (Yes/No), specialisation (Mkt&HR/Mkt&Fin), status(Placed/ Not Placed) - target variable.
- **Numerical Columns:** ssc_p, hsc_p, degree_p, etest_p, mba_p

Exploratory Data Analysis(EDA)

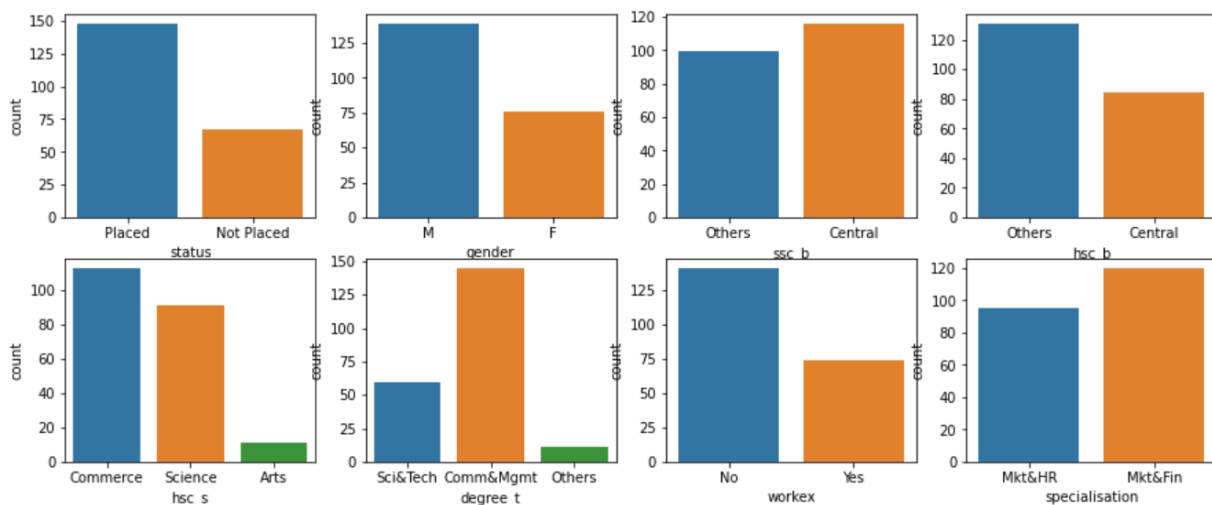
Correlation matrix:

```
cor=data.corr()  
plt.figure(figsize=(5,3))  
sns.heatmap(cor,annot=True)
```



Univariate Analysis

```
fig,ax=plt.subplots(2,4,figsize=(15,6))
sns.countplot(data.status, ax=ax[0][0])
sns.countplot(data.gender, ax=ax[0][1])
sns.countplot(data.ssc_b, ax=ax[0][2])
sns.countplot(data.hsc_b, ax=ax[0][3])
sns.countplot(data.hsc_s, ax=ax[1][0])
sns.countplot(data.degree_t, ax=ax[1][1])
sns.countplot(data.workex, ax=ax[1][2])
sns.countplot(data.specialisation, ax=ax[1][3])
```

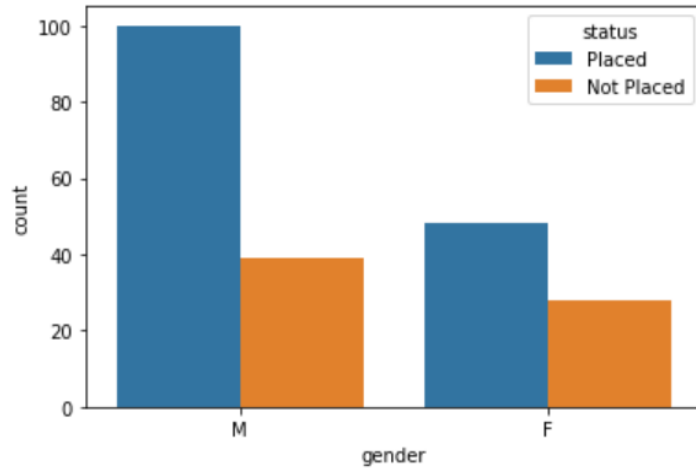


Observations:

- Most of our candidates(~68%) from our dataset **got placed** in a company
- We have **more male**(~65%) candidates than female
- We have candidates who did **commerce** as their hsc course and as well as undergrad
- **Science background** candidates are the second highest in both the cases
- Candidates from **Marketing and Finance** dual specialization are high
- Most of our candidates from our dataset **don't have any work experience**

Effect of Gender on Placement Status

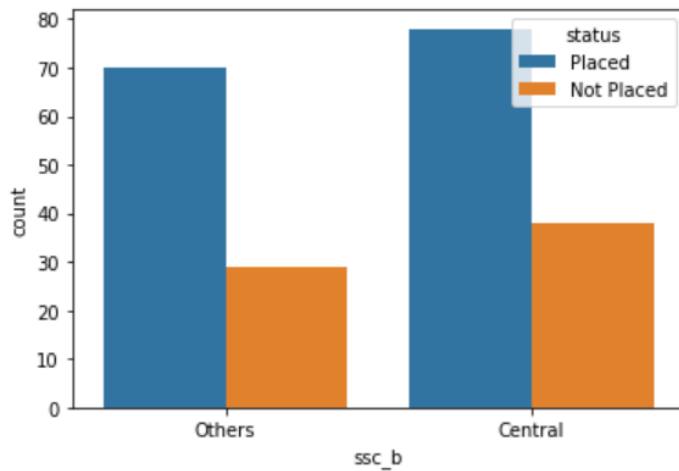
```
sns.countplot('gender',hue='status',data=data)
plt.show()
```



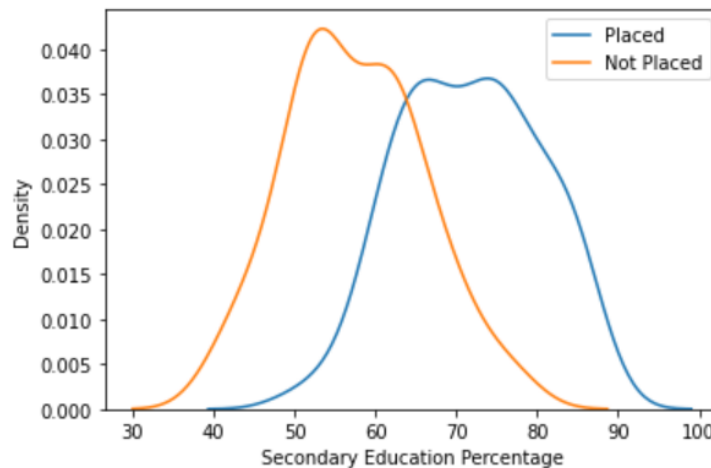
- There are around 68% of males and more male candidates get placed but both types of genders have an approximately equal probability of getting placed.

Effect of Secondary Education(10th) on Placement Status

```
sns.countplot('ssc_b', hue='status', data=data)  
plt.show()
```



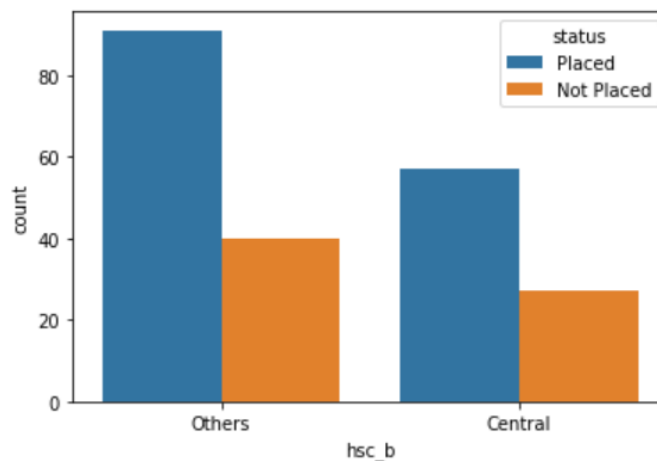
```
sns.kdeplot(data.ssc_p[data.status=='Placed'])
sns.kdeplot(data.ssc_p[data.status=='Not Placed'])
plt.legend(['Placed', 'Not Placed'])
plt.xlabel('Secondary Education Percentage')
plt.show()
```



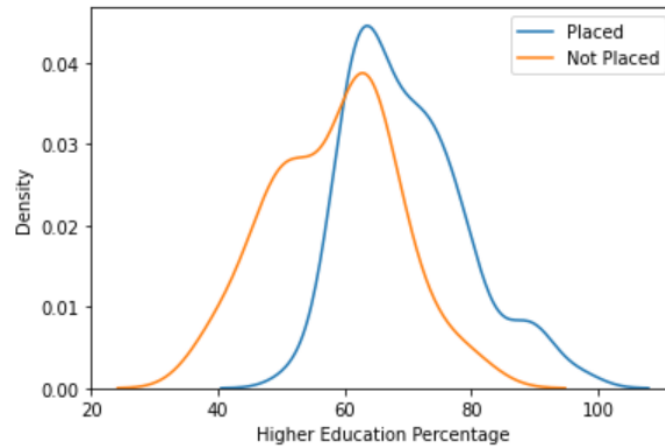
- We see that there are more number of students from the central board compared to all other boards and the board of education doesn't have a significant effect on the placement status
- We see that students with higher secondary education percentage above 65% have better chances of getting placed
- Minimum percentage for a student to get placed is 50%

Effect of Higher Education(12th) on Placement Status

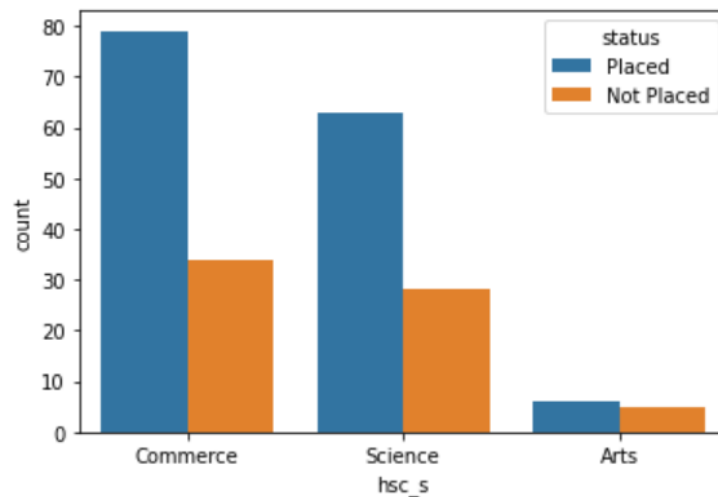
```
sns.countplot('hsc_b', hue='status', data=data)
plt.show()
```



```
sns.kdeplot(data.hsc_p[data.status=='Placed'])
sns.kdeplot(data.hsc_p[data.status=='Not Placed'])
plt.legend(['Placed', 'Not Placed'])
plt.xlabel('Higher Education Percentage')
plt.show()
```



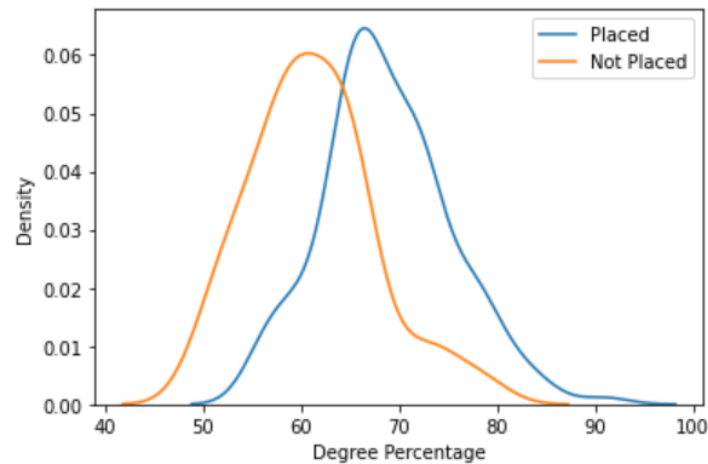
```
sns.countplot('hsc_s', hue='status', data=data)
plt.show()
```



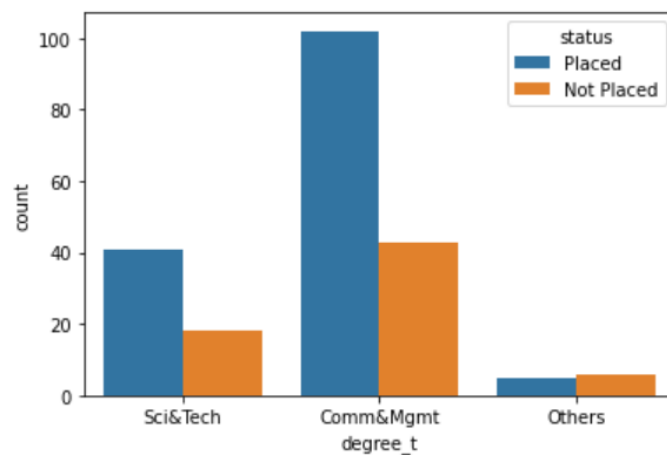
- Students having percentage above 70% get more placements while the number of students placed with percentage below 60% drops steeply
- Minimum higher education percentage for a student to get placed is 50%
- Most of the students are from other boards and the board of higher education doesn't effect the probability of getting placed
- We see that the most common branch is Commerce and least common branch is Arts
- Approximately 69% of the total commerce students, 71% of total science students are placed and 45% of total arts students are not placed
- We conclude that science and commerce branch students have better chance at getting placed than arts students

Effect of Degree on Placement Status

```
sns.kdeplot(data.degree_p[data.status=='Placed'])
sns.kdeplot(data.degree_p[data.status=='Not Placed'])
plt.legend(['Placed', 'Not Placed'])
plt.xlabel('Degree Percentage')
plt.show()
```



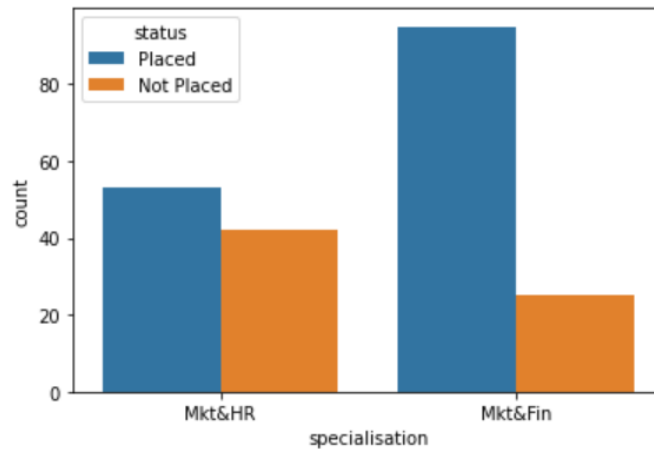
```
sns.countplot('degree_t', hue='status', data=data)
plt.show()
```



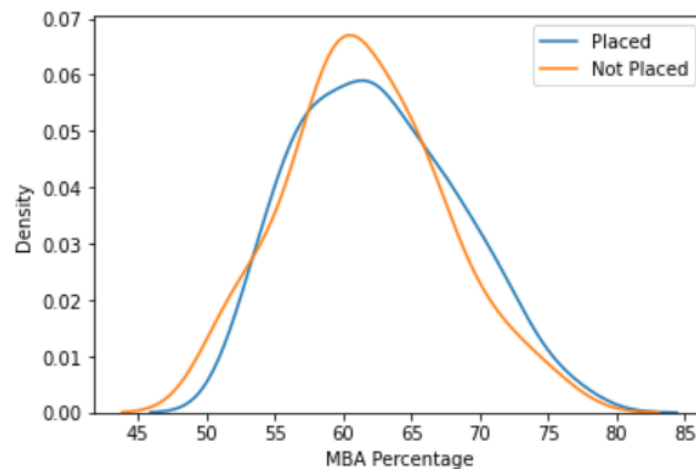
- Most of the students have opted for Commerce and Management for their degree
- Students of Science and Commerce are have more probability to be placed when compared to that of the other branches
- More placements for students with percentage above 65%
- Students should have a minimum of 50% in their degree to get placed

Effect of MBA on Placement Status

```
sns.countplot('specialisation', hue='status', data=data)
plt.show()
```

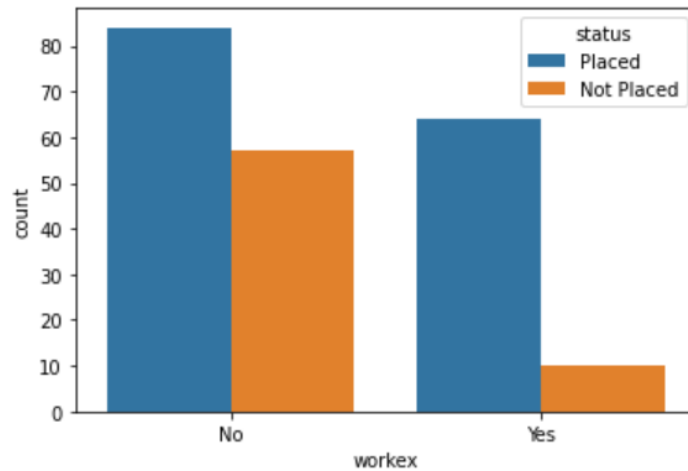
```
sns.kdeplot(data.mba_p[data.status=='Placed'])
sns.kdeplot(data.mba_p[data.status=='Not Placed'])
plt.legend(['Placed', 'Not Placed'])
plt.xlabel('MBA Percentage')
plt.show()
```



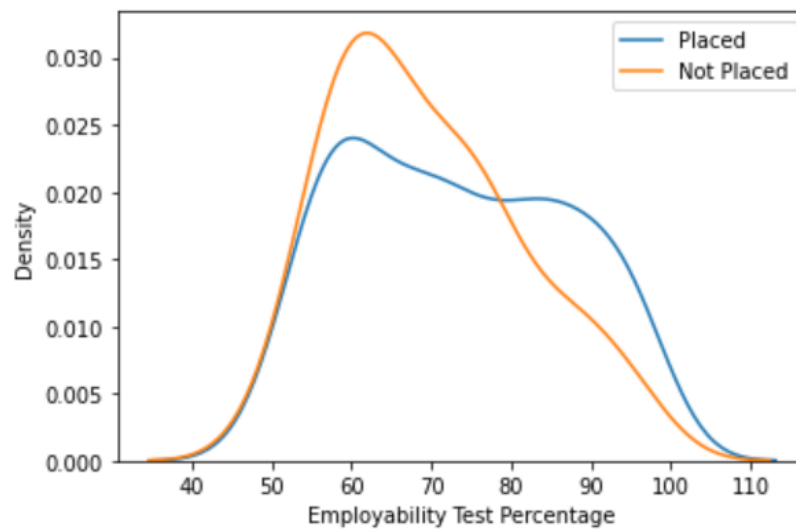
- The placements of Mkt&HR are lower compared to Mkt&Fin
- Better MBA percentage does not guarantee placements

Effect of Work Experience and Employability on Placement Status

```
sns.countplot('workex', hue='status', data=data)
plt.show()
```



```
sns.kdeplot(data.etest_p[data.status=='Placed'])
sns.kdeplot(data.etest_p[data.status=='Not Placed'])
plt.legend(['Placed', 'Not Placed'])
plt.xlabel('Employability Test Percentage')
plt.show()
```



- Students with higher work experience are more likely to get placed
- Employability test percentage greater than 80 have higher probability of getting placed
- The students with 50-75% have lesser chance of getting placed

Feature Encoding

Drop columns which do not add much significance - SSC Board (ssc_b) and HSC Board (hsc_b)

```
data.drop(['ssc_b', 'hsc_b'], axis=1, inplace=True)
```

We then map categorical features to numeric ones:

```
data['gender'] = data.gender.map({'M':0, 'F':1})
data['hsc_s'] = data.hsc_s.map({'Commerce':0, 'Science':1, 'Arts':2})
data['degree_t'] = data.degree_t.map({'Comm&Mgmt':0, 'Sci&Tech':1, 'Others':2})
data['workex'] = data.workex.map({'Yes':0, 'No':1})
data['specialisation'] = data.specialisation.map({'Mkt&HR':0, 'Mkt&Fin':1})
data['status'] = data.status.map({'Placed':1, 'Not Placed':0})
```

Machine Learning model

Creating input variables(X) and target variables(y) from data

```
X = data.copy().drop('status', axis=1)
y = data['status']
```

Using Standard scaler to scale the features

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X = scaler.fit_transform(X)
```

Using train test split on the data

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.1)
```

Using ML Algorithms:

Multiple algorithms were used for training purposes like Decision Tree, Random Forest, SVC, Logistic Regression, K-nearest Neighbors, etc.

Among all these algorithms, **Logistic Regression** performed the best and gave an accuracy score of **90%** when used on test data.

```
from sklearn.metrics import accuracy_score, roc_auc_score
from sklearn.linear_model import LogisticRegression
clf = LogisticRegression()
clf.fit(X_train, y_train)
predictions = clf.predict(X_test)
score = accuracy_score(y_test, predictions)
score
```

0.9090909090909091

Conclusion

From the analysis report on Campus Recruitment dataset here are my following conclusions

- **Educational percentages and Past work experience** are highly influential for a candidate to get placed
- There are **no gender discrimination** while hiring
- The board of education of the candidate did not have any significance
- Majority of the students chose Commerce branch for their degree and higher education
- **Logistic Regression** worked better than all other Ensemble models by giving an accuracy of **90%**